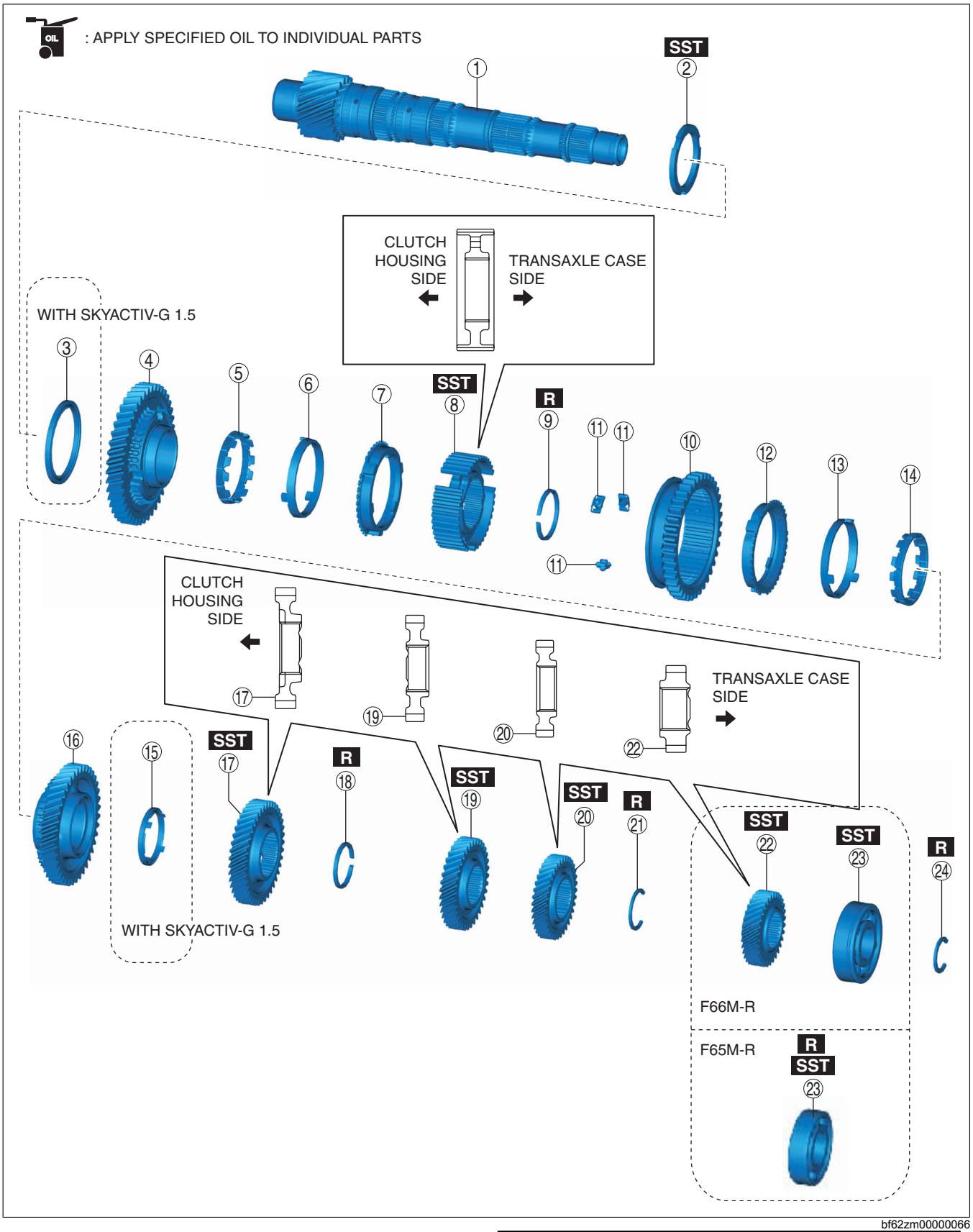
SECONDARY SHAFT COMPONENT ASSEMBLY

id051500169300

Note
- The thrust washer cannot be removed from the secondary shaft.
- If the secondary shaft is replaced, it is necessary to assemble the thrust washer to the secondary shaft.

1. Assemble the secondary shaft component in the order shown in the figure.



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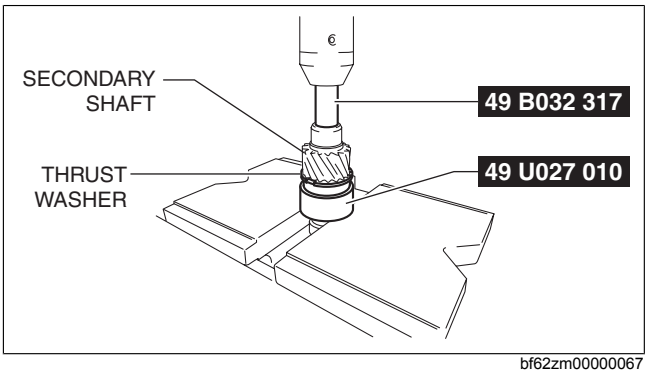
1	Secondary shaft
2	Thrust washer (See Thrust Washer Assembly Note.)
3	Friction damper (secondary 1st gear) (with SKYACTIV-G 1.5)
4	Secondary 1st gear

5	Inner cone ring (secondary 1st gear)
6	Middle cone ring (secondary 1st gear) (See Middle Cone Ring (Secondary 1st Gear) Assembly Note.)
7	Synchronizer ring (secondary 1st gear) (See Synchronizer Ring (Secondary 1st Gear) Assembly Note.)

8	Clutch hub (See Clutch Hub Assembly Note.)
9	Clip
10	Clutch hub sleeve (See Clutch Hub Sleeve Assembly Note.)
11	Synchronizer key (See Synchronizer Key Assembly Note.)
12	Synchronizer ring (secondary 2nd gear) (See Secondary 3rd Gear, Friction Damper, Secondary 2nd Gear, Inner Cone Ring, Middle Cone Ring, Synchronizer Ring Assembly Note.)
13	Middle cone ring (secondary 2nd gear) (See Secondary 3rd Gear, Friction Damper, Secondary 2nd Gear, Inner Cone Ring, Middle Cone Ring, Synchronizer Ring Assembly Note.)
14	Inner cone ring (secondary 2nd gear) (See Secondary 3rd Gear, Friction Damper, Secondary 2nd Gear, Inner Cone Ring, Middle Cone Ring, Synchronizer Ring Assembly Note.)
15	Friction damper (secondary 2nd gear) (with SKYACTIV-G 1.5) (See Secondary 3rd Gear, Friction Damper, Secondary 2nd Gear, Inner Cone Ring, Middle Cone Ring, Synchronizer Ring Assembly Note.)
16	Secondary 2nd gear (See Secondary 3rd Gear, Friction Damper, Secondary 2nd Gear, Inner Cone Ring, Middle Cone Ring, Synchronizer Ring Assembly Note.)
17	Secondary 3rd gear (See Secondary 3rd Gear, Friction Damper, Secondary 2nd Gear, Inner Cone Ring, Middle Cone Ring, Synchronizer Ring Assembly Note.)
18	Clip
19	Secondary 4th gear (See Secondary 4th Gear Assembly Note.)
20	Secondary 5th gear (See Secondary 5th Gear Assembly Note.)
21	Clip
22	Secondary 6th gear (See Secondary 6th Gear Assembly Note.)
23	Ball bearing (See Ball Bearing Assembly Note.)
24	Clip

Thrust Washer Assembly Note

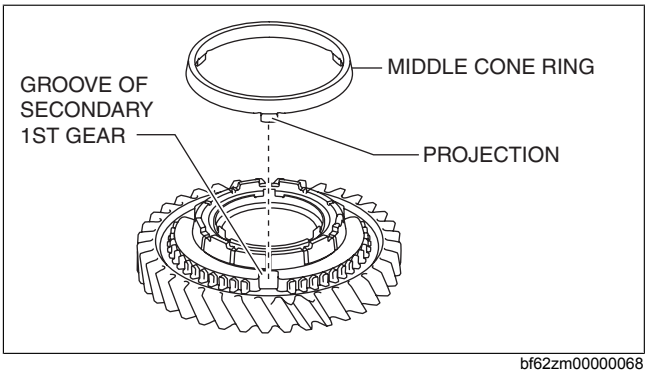
1. Set the **SSTs**, thrust washer, and secondary shaft in the press as shown in the figure.
2. Assemble the thrust washer using a press.



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Middle Cone Ring (Secondary 1st Gear) Assembly Note

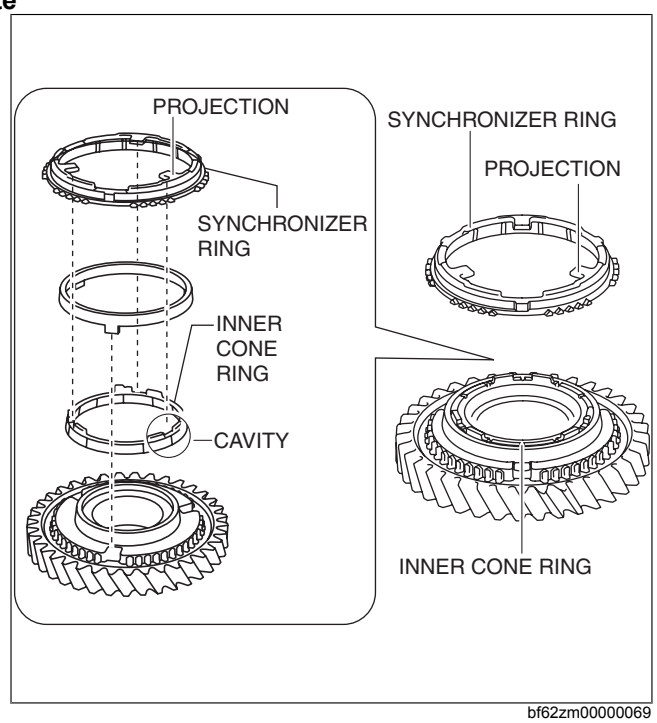
1. Assemble the middle cone ring so that the middle cone ring projection and the groove of the secondary 1st gear are engaged as shown in the figure.



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Synchronizer Ring (Secondary 1st Gear) Assembly Note

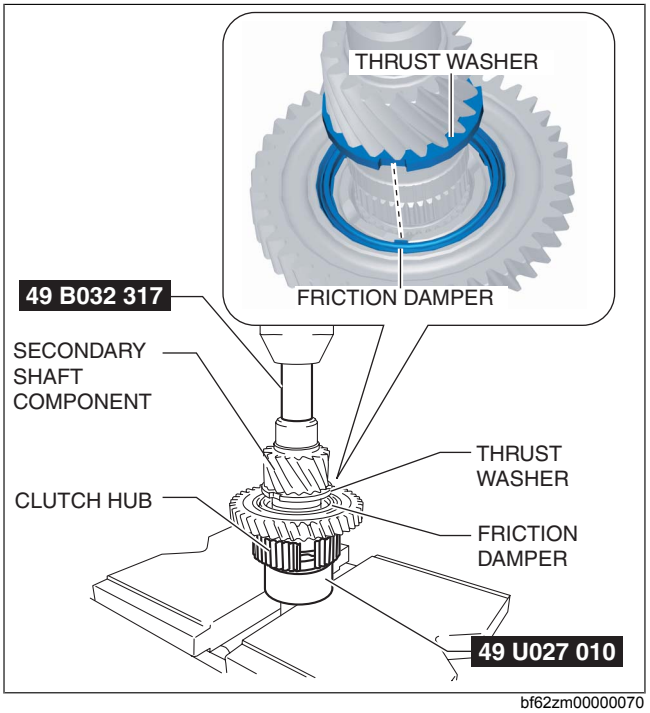
1. Assemble the synchronizer ring so that the synchronizer ring projection and the cavity of the inner cone ring are engaged as shown in the figure.



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Clutch Hub Assembly Note

1. Set the **SSTs**, clutch hub, and secondary shaft component in a press as shown in the figure.
2. Align the projection of the synchronizer ring with the groove of the clutch hub key.
3. Align the projection of the friction damper with the groove of the thrust washer.
4. Assemble the clutch hub using the press.

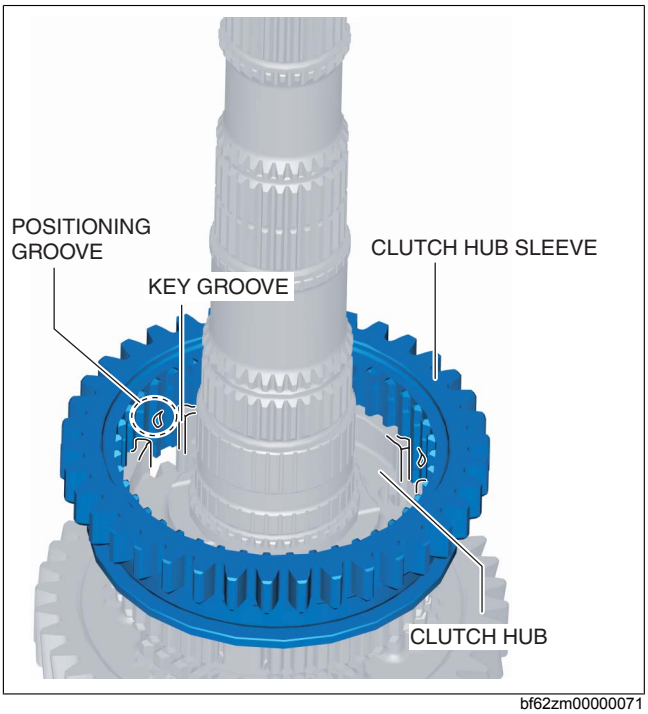


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Clutch Hub Sleeve Assembly Note

1. Align the positioning groove of the clutch hub sleeve with the center of the clutch hub key groove as shown in the figure, and assemble the clutch hub sleeve.

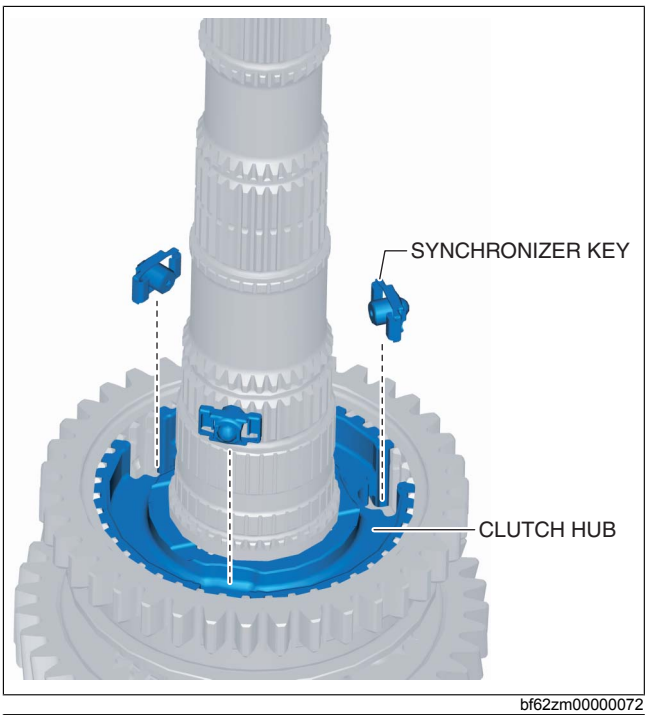
   Note
   - A groove is cut across the entire circumference of the clutch hub sleeve, however the deeper grooves in 3 locations are the reference position grooves.



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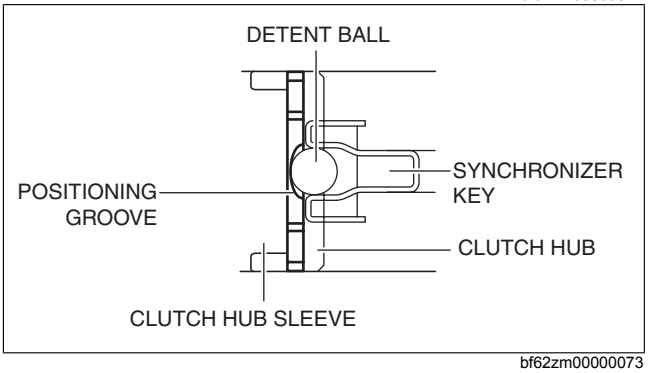
Synchronizer Key Assembly Note

1. Align the synchronizer keys with the clutch hub grooves as shown in the figure and assemble.



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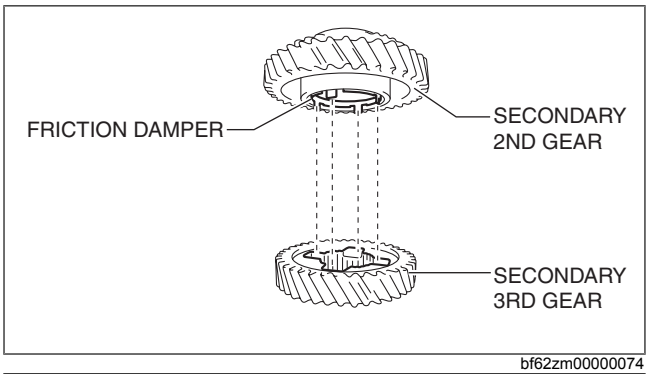
2. After the synchronizer key is assembled, verify that the detent ball of the synchronizer key is engaged with the positioning groove of the clutch hub sleeve.



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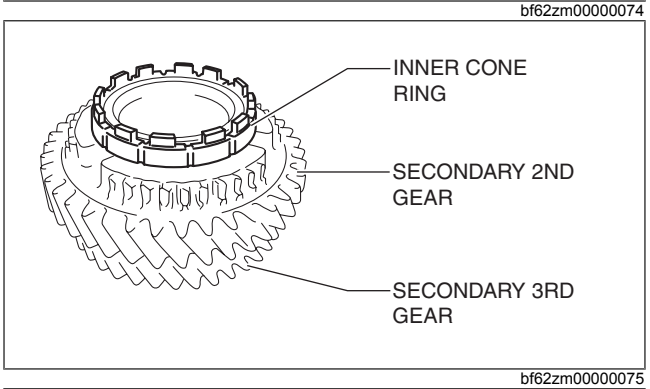
Secondary 3rd Gear, Friction Damper, Secondary 2nd Gear, Inner Cone Ring, Middle Cone Ring, Synchronizer Ring Assembly Note

1. Assemble the secondary 2nd gear so that the projection of the friction damper assembled to the secondary 2nd gear is engaged with the groove of the secondary 3rd gear as shown in the figure.



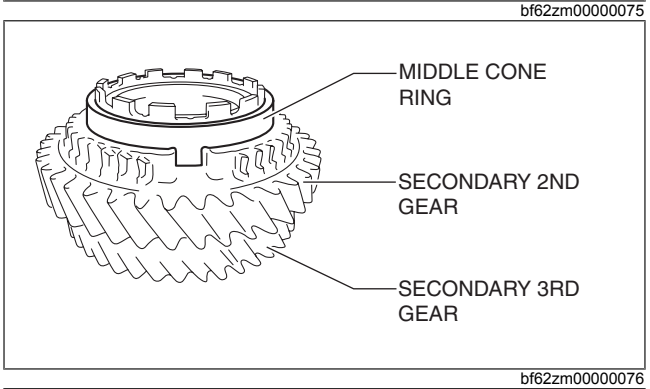
bf62zm00000074

2. Assemble the inner cone ring to the secondary 2nd gear.



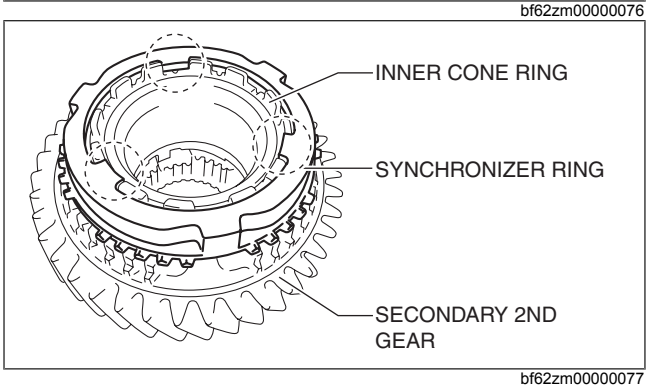
bf62zm00000075

3. Assemble the middle cone ring so that the middle cone ring projection and the groove of the secondary 2nd gear are engaged as shown in the figure.



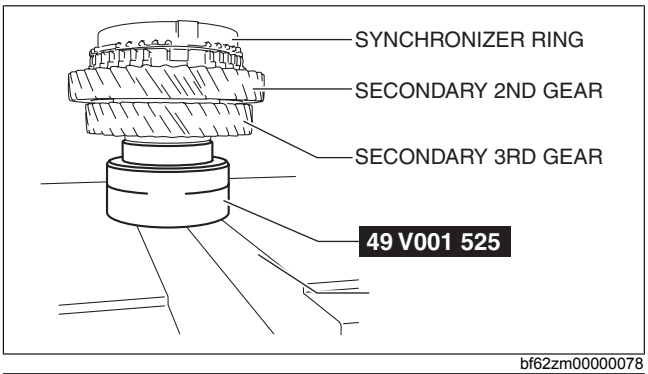
bf62zm00000076

4. Assemble the synchronizer ring so that the synchronizer ring projections and the cavities of the inner cone ring are engaged as shown in the figure.



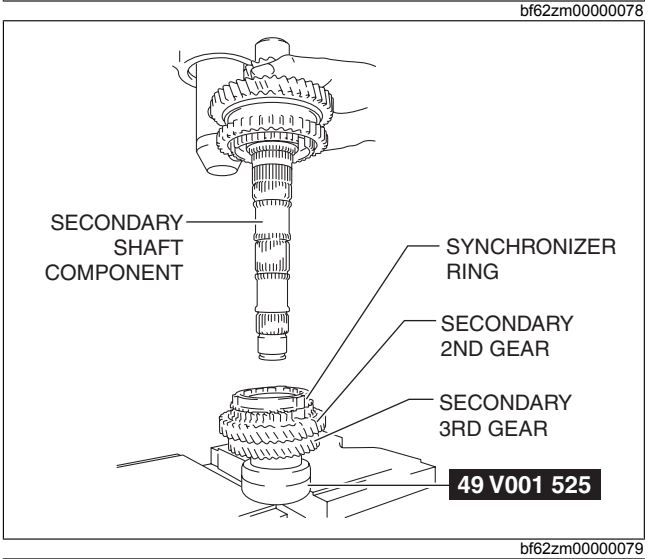
bf62zm00000077

5. Set the **SST** and the parts from the assembled secondary 3rd gear to the synchronizer ring in a press as shown in the figure.



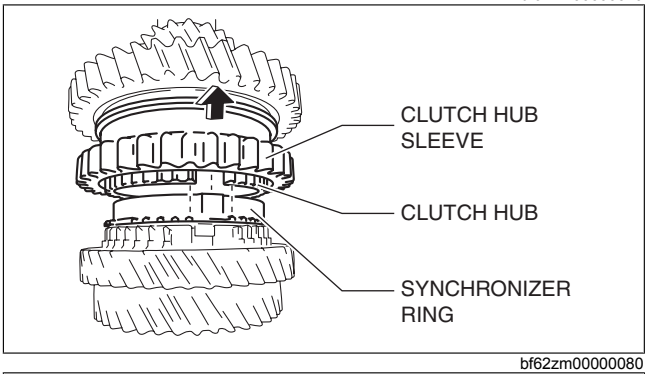
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6. Set the secondary shaft component in a press as shown in the figure.



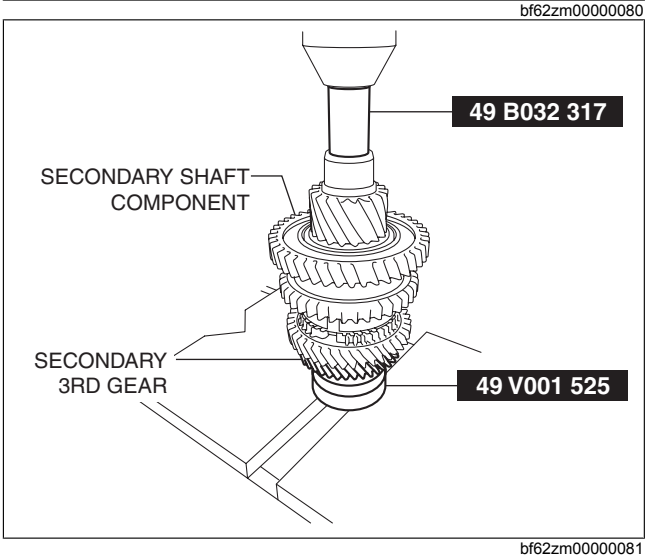
bf62zm00000079

7. Slide the clutch hub sleeve as shown in the figure so that the groove of the clutch hub is visible.
8. Align the projection of the synchronizer ring with the groove of the clutch hub key.



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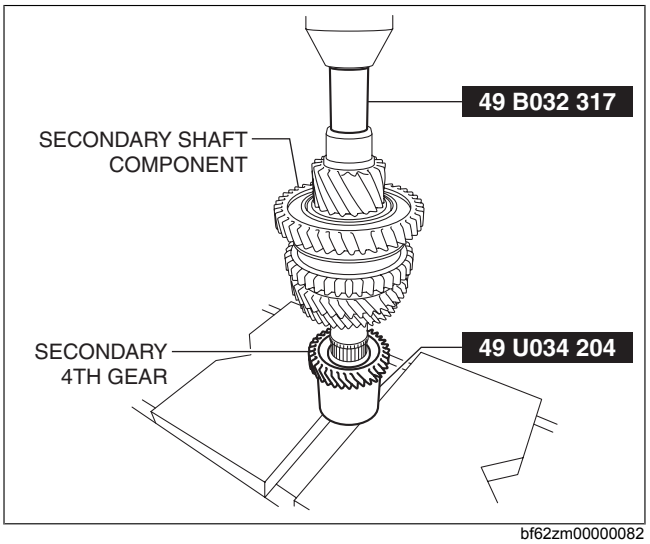
9. Set the secondary shaft component and the **SSTs** in the press as shown in the figure.
10. Assemble the parts from the secondary 3rd gear to the synchronizer ring using the press.



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Secondary 4th Gear Assembly Note

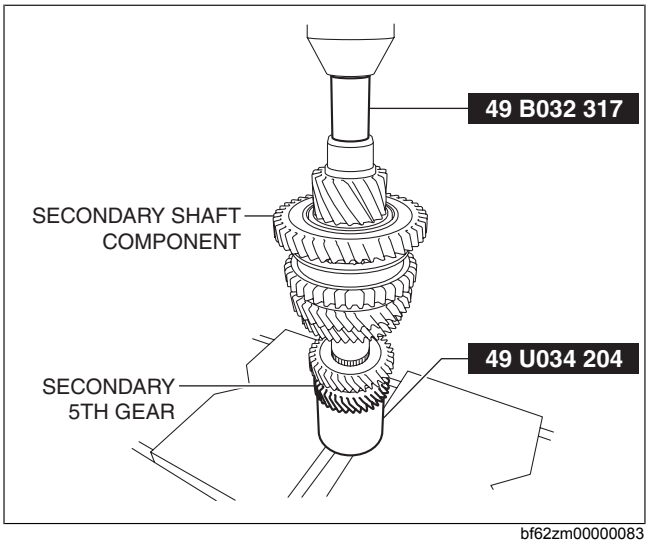
1. Set the **SSTs**, secondary 4th gear, and secondary shaft component in the press as shown in the figure.
2. Assemble the secondary 4th gear using the press.



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Secondary 5th Gear Assembly Note

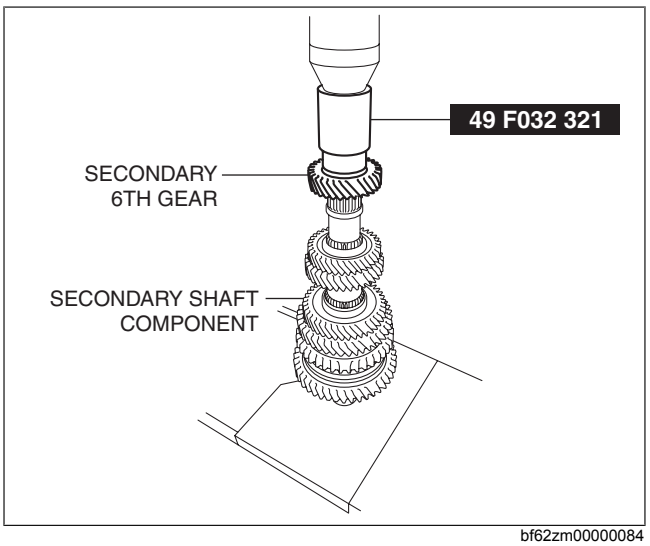
1. Set the **SSTs**, secondary 5th gear, and secondary shaft component in the press as shown in the figure.
2. Assemble the secondary 5th gear using the press.



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Secondary 6th Gear Assembly Note

1. Set the secondary shaft component, secondary 6th gear, and the **SST** in a press as shown in the figure.
2. Assemble the secondary 6th gear using the press.

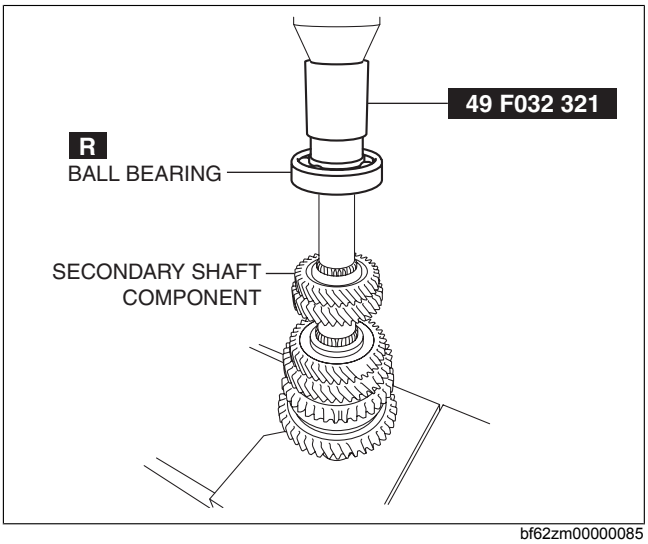


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Ball Bearing Assembly Note

1. Set the secondary shaft component, new ball bearings, and **SSTs** in a press as shown in the figure.

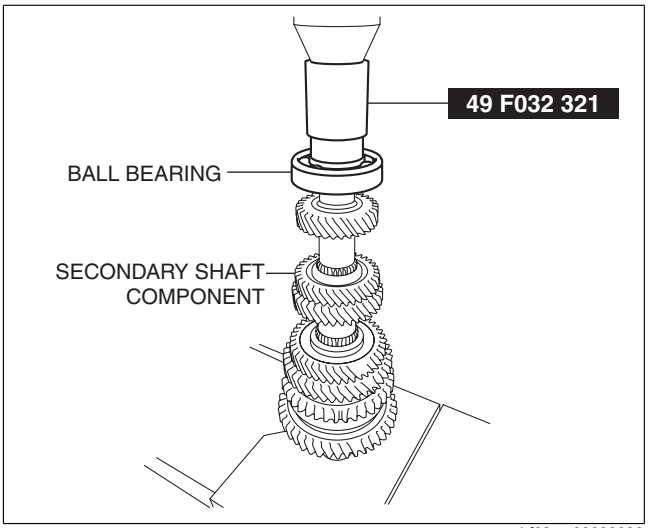
F65M-R



bf62zm00000085

F66M-R

2. Assemble the ball bearings using the press.
3. Assemble a new clip.



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